

Dynamic Factor Model with Mean-Variance Optimisation

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Abstract

This paper develops a unified quantitative equity investment framework combining a dynamic cross-sectional factor model with a statistical risk engine and constrained portfolio optimisation. Five economically motivated factors—market beta, momentum, size, low volatility, and long-term reversal—are constructed from cross-sectional equity data and transformed through winsorisation, standardisation, and sequential orthogonalisation.

Predictive signals are dynamically weighted using rolling Information Coefficients (IC), allowing the model to adapt to changing market regimes. In parallel, a Principal Component Analysis (PCA) framework is used to estimate a low-dimensional representation of the covariance structure of returns. A residual covariance component is derived from the difference between empirical and PCA-reconstructed covariance matrices.

Portfolio weights are constructed using risk-adjusted inverse covariance scaling and further refined through mean-variance optimisation under leverage and market-neutrality constraints. The framework is evaluated through backtesting and benchmark comparison.

1 Introduction

Equity returns exhibit strong cross-sectional structure driven by systematic risk factors and behavioural anomalies. Classical factor models such as those of Fama and French provide interpretable decompositions of expected returns, while modern quantitative approaches increasingly integrate statistical learning techniques and dynamic weighting schemes.

This paper proposes a modular framework that separates the investment process into three components:

1. **Alpha model:** construction and combination of cross-sectional predictive signals
2. **Risk model:** estimation of covariance structure using PCA
3. **Portfolio construction:** constrained optimisation using risk-adjusted signals

The objective is to build a robust and adaptable pipeline that improves stability in high-dimensional asset universes while preserving economic interpretability.

2 Theoretical Framework

2.1 Cross-Sectional Factor Model

Asset returns are modelled as:

$$r_{i,t} = \sum_{k=1}^K \beta_{i,k,t} f_{k,t} + \epsilon_{i,t}, \quad (1)$$

where $r_{i,t}$ is the return of asset i at time t , $\beta_{i,k,t}$ is the factor exposure, $f_{k,t}$ denotes factor returns, and $\epsilon_{i,t}$ is an idiosyncratic error term.

2.2 Factor Signal Construction

For each factor $k \in 1, \dots, 5$, raw signals $s_{i,t}^{(k)}$ are constructed and processed as follows:

Winsorisation:

$$s_{i,t}^{(k)} \leftarrow \text{Winsorise}(s_{i,t}^{(k)}) \quad (2)$$

Cross-sectional standardisation:

$$\tilde{s}_{i,t}^{(k)} = \frac{s_{i,t}^{(k)} - \mu^{(k)}_t}{\sigma^{(k)}_t} \quad (3)$$

Sequential orthogonalisation: Each factor is residualised against previously defined factors:

$$E[\tilde{s}_{i,t}^{(k)} \mid \tilde{s}_{i,t}^{(1)}] \quad (4)$$

2.3 Information Coefficient Weighting

Predictive power is measured via the Information Coefficient:

$$IC^{(k)}_t = \text{corr}(\tilde{s}_{i,t}^{(k)}, r_{i,t+1}) \quad (5)$$

Factor weights are defined as:

$$w_t^{(k)} \propto IC_t^{(k)} \quad (6)$$

providing a dynamic weighting mechanism that adapts over time.

3 PCA-Based Risk Model

3.1 PCA Decomposition

Let Σ_{sample} denote the sample covariance matrix of asset returns computed over a rolling window.

The covariance matrix is decomposed as:

$$\Sigma_{\text{sample}} = B \Lambda B^\top \quad (7)$$

where B is the eigenvector matrix and Λ is the diagonal matrix of eigenvalues.

3.2 Low-Rank Approximation

The number of principal components (PCs) k is selected such that

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^N \lambda_j} \geq 0.80 \quad (8)$$

The PCA covariance estimator is:

$$\Sigma_{\text{PCA}} = B_k \Lambda_k B_k^\top \quad (9)$$

3.3 Residual Risk

Residual Covariance is defined as:

$$D = \Sigma_{\text{sample}} - \Sigma_{\text{PCA}} \quad (10)$$

This term captures idiosyncratic covariance structure.

4 Portfolio Construction

4.1 Long-Short Quantile Portfolio Construction

As a benchmark for evaluating the predictive power of the cross-sectional factor model, a long-short portfolio is constructed based on ranked aggregated factor scores. At each rebalancing period t , assets are sorted according to their composite signal $s_{i,t}$, formed from the combination of standardised and orthogonalised factor exposures.

The investment universe is partitioned into two disjoint sets:

- Long leg: top 20% of assets ranked by $s_{i,t}$
- Short leg: bottom 20% of assets ranked by $s_{i,t}$

Let $\mathcal{L}_t$ and $\mathcal{S}_t$ denote the sets of long and short positions respectively, with equal cardinality:

$$n_L = n_S = n$$

where n corresponds to the number of assets selected in each tail (e.g. for 60 assets total, $n = 12$).

Formally:

$$\mathcal{L}_t = \{i \mid s_{i,t} \in Q_{80-100}\}, \quad \mathcal{S}_t = \{i \mid s_{i,t} \in Q_{0-20}\}$$

where Q_{a-b} denotes the cross-sectional quantile range at time t .

Portfolio weights are assigned uniformly across each leg. Each long position receives weight $+\frac{1}{2n}$, while each short position receives weight $-\frac{1}{2n}$:

$$w_{i,t} = \begin{cases} \frac{1}{2n}, & i \in \mathcal{L}_t \\ -\frac{1}{2n}, & i \in \mathcal{S}_t \\ 0, & \text{otherwise} \end{cases}$$

This construction ensures exact dollar neutrality:

$$\sum_i w_{i,t} = 0$$

and unit gross exposure:

$$\sum_i |w_{i,t}| = 1$$

The resulting portfolio isolates the cross-sectional predictive power of the factor signals prior to any risk adjustment or optimisation procedure.

4.2 Risk-Adjusted Allocation

Let s_t denote the aggregated factor score vector.

Normalisation:

$$\tilde{s}_t = \frac{s_t}{\sum_i s_{i,t}} \quad (11)$$

Risk adjustment:

$$w_t = D^{-1} \tilde{s}_t \quad (12)$$

Portfolio constraints: Final weights are normalised such that:

$$\sum_i |w_{i,t}| = 1, \quad \sum_i w_{i,t} = 0 \quad (13)$$

4.3 Mean-Variance Optimisation

A classical optimisation problem is solved:

$$\max_w w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w \quad (14)$$

where μ is the expected return vector derived from factor signals and Σ is the PCA-based covariance estimator.

Varying λ generates an efficient frontier.

5 Methodology

5.1 Data

Equity monthly closing data is sourced from yfinance and transformed into a clean cross-sectional return panel.

5.2 Pipeline

- Construct factor signals (beta, momentum, size, volatility, reversal)
- Winsorise, standardise, and orthogonalise signals

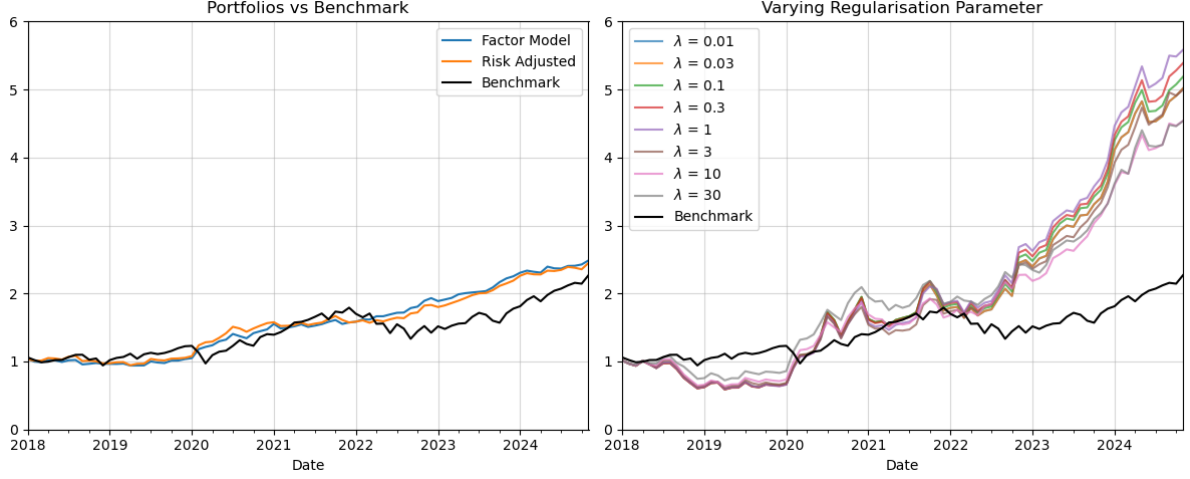


Figure 1: Enter Caption

- Compute rolling IC and construct dynamic factor weights
- Estimate covariance using rolling PCA (36-period window)
- Construct risk-adjusted portfolio using inverse covariance weighting
- Solve mean-variance optimisation problem
- Evaluate performance via backtesting

6 Results

	Factor Model	Risk-Adjusted Model	Mean-Variance ($\lambda = 1$)
Cumulative PnL	149%	145%	458%
Sharpe Ratio	1.49	1.35	0.94
Max Drawdown	-9%	-13%	-41%

Table 1: Performance comparison of portfolio construction methods.

Information ratio IC and ICIR of factor signals Comparison against benchmark portfolios Efficient frontier under varying λ

7 Conclusion

This paper presents a modular quantitative investment framework integrating dynamic cross-sectional factor modelling, PCA-based covariance estimation, and constrained portfolio optimisation.

The separation between alpha generation and risk modelling enhances interpretability and robustness. Dynamic IC-based weighting improves adaptability across market regimes, while PCA provides a stable low-dimensional representation of covariance structure in high-dimensional asset spaces.

Future extensions include transaction cost modelling, sector neutrality constraints, and time-varying covariance models such as DCC-GARCH.

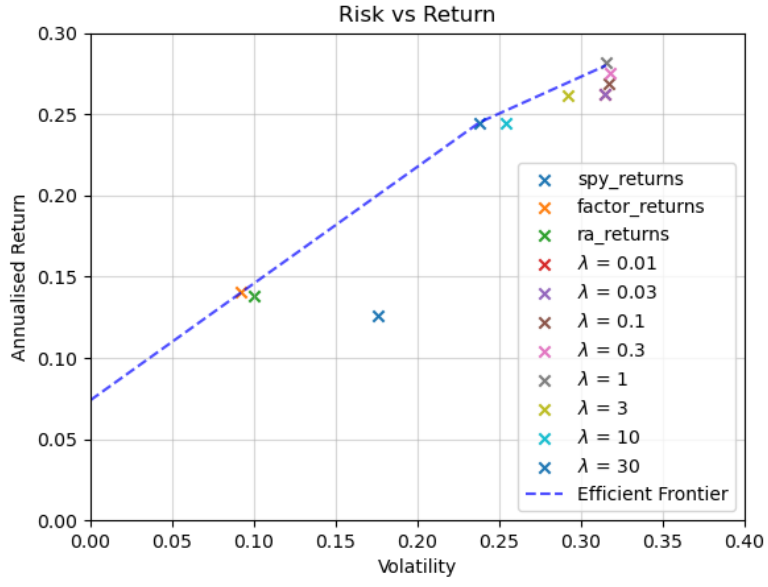


Figure 2: Enter Caption

References

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